Hussain Singapuri

PYTHON / AI DEVELOPER

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PROFESSIONAL SUMMARY

Python / AI Developer with 3 years of professional full-stack experience at a US-based medical-device company, building web applications, a **CRM** and internal **AI agents**. Builds **REST APIs** in Core **Python (FastAPI, asyncio)** on **PostgreSQL / SQL**, and integrates **LLM** and **Generative AI** models (Claude, Gemini) into agentic workflows with RAG, tool use, evals and guardrails. Uses **Git** version control, GitHub Actions and Docker daily, with AI application deployment on cloud platforms (Vercel, Supabase, Neon), and ships fast with **Claude Code**. Claude Certified Developer.

CORE COMPETENCIES

Core Python | FastAPI & REST APIs | SQL & PostgreSQL | LLM Integration | Generative AI | AI Agents & Agentic Workflows | RAG & Retrieval | Git & GitHub Actions

WORK EXPERIENCE

NeuroNetworks AI Ed

2026 - Present

CTO & Principal Course Architect

Pune, India

- Built and run the website neuronetwrks.com with a Python static-site generator deployed on Vercel, including programmatic SEO
- Lead technology for an Indian AI and tech education brand (Neuronetworks AI Ed Private Limited) offering programmes for students after Class 10
- Architected the curriculum: a 1-year AI Foundation course and five 3-year specialisations (AI/ML Engineering, Software Development, Data Science & AI, Product & UX / AI Design, Cybersecurity)
- Designing Neuromind Practical AI Lab, a browser-based IDE and sandbox with an AI tutor: phased spec, server sizing, engineering team plan and CAPEX/OPEX model

US-based medical-device company

Mar 2023 - Sep 2026

Full Stack Developer

Remote

- Built full-stack web applications and several apps, including a **CRM**, for the company
- Built **AI agents** for internal use
- Performed penetration / security testing on 5+ of the company's mobile apps, all live on the Google Play Store

PROJECTS

multi-agent-research-crew

Open Source

Agentic workflow in Python (asyncio, Claude): planner, parallel researcher, writer and verifier agents on a custom DAG orchestrator with retries, timeouts, token budgets and claim-level citation checking.

Python, asyncio, Claude

site-to-agent

Open Source

LLM integration and RAG: crawler, hybrid BM25 + vector retrieval with RRF (pgvector), cited answers, SSRF-hardened fetching and an eval harness.

Next.js, TypeScript, pgvector

lead-intel-engine

Open Source

Python / FastAPI service with SQLite that discovers, enriches and scores local-business leads with a robots-aware, rate-limited crawler and

transparent scoring.

Python, FastAPI, SQLite

ai-lead-receptionist Open Source

AI receptionist for home-service businesses: missed-call text-back, SMS and web chat that qualifies leads and books jobs, owner dashboard.

Next.js, TypeScript, Twilio, Claude, Supabase

PageForge Portfolio project

Guardrails for LLM output: AI drafts only from approved facts, a deterministic validator flags unsupported numbers, a reviewer signs off claim by claim.

Next.js 16, React 19, Tailwind v4, Sanity, Supabase, Gemini, PostHog, HubSpot

Orbital Surplus Portfolio project, fictional brand

PostgreSQL transactions on a money path: race-safe stock holds in one transaction, idempotent payment confirmation via signed Stripe webhook (test mode); 9 end-to-end API tests.

three.js, GSAP, WebXR, Node 20, Postgres (Neon), Stripe, Resend

nanoserve Open Source

LLM inference engine from scratch: paged KV cache, continuous batching, speculative decoding, OpenAI-compatible streaming API; 3.1x throughput over sequential decoding.

Python, PyTorch, FastAPI

Personal portfolio with full-stack backend Portfolio project

Rate-limited enquiry form, booking with paid Stripe sessions (slot holds, signed webhooks), HMAC-signed CRM events, nonce-based CSP; 10 end-to-end API tests.

Vanilla ES modules, GSAP, Node 20, Postgres (Neon), Stripe, Resend

More projects

Nightjar & Co., contractor-ai-ops-dashboard, Safey Inventory, TripSync, Lumen, Kilnwood Roasters, Cairnmoor, Lowlight Video Club

EDUCATION

Bachelor's, Computer Science (AI & Machine Learning) Vishwakarma Institute of Technology, Pune 2022-2025

Pune

CERTIFICATIONS

Claude Certified Developer	Anthropic, Foundations (CCDV-F)
Claude Certified Orchestrator	Anthropic, Foundations (CCAO-F)
Generative AI Leader	Google Cloud
Claude Code in Action	Anthropic
Claude Code 101	Anthropic

SKILLS

AI / LLM: Claude API, Gemini API, AI agents, multi-agent orchestration, tool use / function calling, RAG (hybrid BM25 + vector, RRF), evals, prompt-injection defence, LLM inference serving, Claude Code and AI coding agents

Data: PostgreSQL, Neon, Supabase (RLS, Edge Functions), PGlite, pgvector, SQLite, Sanity CMS

Languages: TypeScript, JavaScript, Python, SQL, Dart, HTML, CSS

Back end: Node.js, FastAPI, asyncio, REST APIs, webhooks (HMAC signing), auth (scrypt, sessions, throttling), Stripe Checkout and refunds, Resend

Front end: React 19, Next.js, TypeScript, JavaScript, Tailwind CSS, shadcn/ui, TanStack Start / Query, GSAP, Lenis, three.js, WebGL, Canvas 2D, responsive and accessible UI, motion and interaction design

DevOps: Git (version control), GitHub Actions, Docker, Vercel, AI application deployment on cloud platforms (Vercel, Supabase, Neon)

Security: mobile app penetration testing, CSP and security headers, SSRF hardening, input validation

Automation & integrations: workflow automation, CRM integrations (HubSpot), PostHog, Twilio SMS, web crawling / scraping

Mobile: Flutter, Riverpod, Firebase **Design:** UI/UX design, design systems, technical SEO, JSON-LD structured data, programmatic SEO